Lecture 9 - Conditional probability

Oct 5, 2026

Def: $P(A|B) = P(A \cap B) / P(B)$, when $P(B) > 0$

read as: prob of A given B

Ex: roll a fair die. A = even, B = at least 4

$P(A \cap B) = 2/6$, $P(B) = 3/6$

so $P(A|B) = 2/3$

Note: $P(A|B)$ is not $P(B|A)$ in general!

Multiplication rule

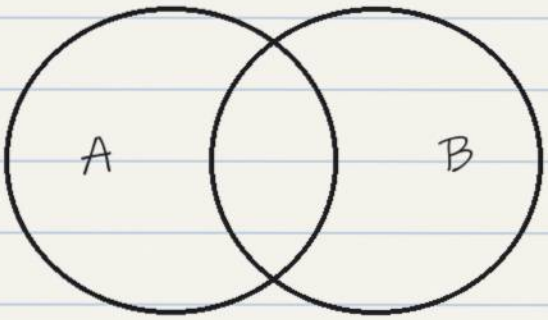
$P(A \cap B) = P(A)\,P(B|A)$

Ex: 2 cards, no replacement

$P(\text{both aces}) = 4/52 \times 3/51 = 1/221$

Chain: $P(A \cap B \cap C) = P(A)\,P(B|A)\,P(C|A \cap B)$

(sketch: Venn diagram, A and B overlap)



Law of total probability

If B1, ..., Bk partition S then

P(A) = sum of P(A|Bi) P(Bi)

Ex: 3 machines make 50%, 30%, 20% of parts

defect rates 1%, 2%, 3%

P(defect) = .005 + .006 + .006 = .017

Lecture 10 - Bayes' theorem

Oct 7, 2026

$P(Bj|A) = P(A|Bj)\,P(Bj) / \text{sum of } P(A|Bi)\,P(Bi)$

same machines: part is defective,

prob it came from machine 3?

= .006 / .017 = 0.353 (about 35%)

prior 20% -> posterior 35%

Independence

A, B independent iff $P(A \cap B) = P(A)\,P(B)$

equivalently $P(A|B) = P(A)$

mutually exclusive is NOT independent

(unless one has prob 0)

Ex: two coin flips, H on 1st and H on 2nd are indep

Lecture 11 - Discrete random variables

Oct 9, 2026

$X : S \to R$, a function on outcomes

prob function $f(x) = P(X = x)$

$f(x) >= 0$ and sum of $f(x) = 1$

cdf $F(x) = P(X <= x)$, non-decreasing

Binomial distribution

$X \sim Bin(n, p)$: number of successes in n indep trials

$f(x) = \binom{n}{x} p^x (1-p)^{(n-x)}, x = 0..n$

Ex: $n = 10, p = 0.3, P(X = 2) = 45(.09)(.7^8) = 0.233$

Midterm 1 is Thursday Oct 8 - ch 1 to 4

Quiz 2 Fri Oct 30 in tutorial (UTD 105)